Chapter 6. Metrization Theorems and Paracompactness

Local finiteness, Nagata–Smirnov, paracompactness, Smirnov metrization

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Urysohn's theorem characterizes metrizability among second countable spaces. Here we characterize it in general.

6.1 Local finiteness

A collection $\mathcal{A}$ of subsets of X is **locally finite** if every point has a neighbourhood meeting only finitely many members, and **countably locally finite** (σ-locally finite) if it is a countable union of locally finite collections.

Lemma 6.1

If $\mathcal{A}$ is locally finite, so is $\{\overline{A} : A \in \mathcal{A}\}$, and $\overline{\bigcup A} = \bigcup \overline{A}$.

Proof. An open set meeting $\overline{A}$ meets A. If $x \in \overline{\bigcup A}$, a neighbourhood meeting only $A_1, \dots, A_k$ shows $x \in \overline{A_1 \cup \cdots \cup A_k} = \bigcup_{i \le k} \overline{A_i}$. ∎

A collection $\mathcal{B}$ **refines** $\mathcal{A}$ if every member of $\mathcal{B}$ lies in some member of $\mathcal{A}$.

Lemma 6.2

Every open cover of a metrizable space has a countably locally finite open refinement that covers the space.

Proof. Well-order the cover $\mathcal{A}$. For each n and $U \in \mathcal{A}$ let $S_n(U) = \{x : B(x, 1/n) \subset U\}$, $T_n(U) = S_n(U) - \bigcup_{V<U} V$, and $E_n(U)$ its $1/(3n)$-neighbourhood. For fixed n the sets $E_n(U)$ are at mutual distance at least $1/(3n)$, so $\{E_n(U)\}_U$ is locally finite; and every x lies in some $E_n(U)$, with U the first member containing x and n large. ∎

6.2 The Nagata–Smirnov metrization theorem

Theorem 6.3 (Nagata–Smirnov)

X is metrizable iff it is regular and has a countably locally finite basis.

Sketch. Necessity: apply Lemma 6.2 to the covers by $1/m$-balls, $m = 1, 2, \dots$. Sufficiency: (i) such an X is normal, and every closed set is a G_δ (a countable intersection of open sets), which gives, for each open U, a continuous $f \ge 0$ with $U = \{f > 0\}$. (ii) For each basis element B in the n-th locally finite family $\mathcal{B}_n$ choose $f_{n,B} : X \to [0, 1/n]$ positive exactly on B. (iii) The map $F(x) = (f_{n,B}(x))_{n,B}$ into $[0,1]^J$ with the *uniform* metric is an imbedding: local finiteness makes the uniform metric continuous on $F(X)$ (near a point only finitely many coordinates per n are nonzero, and the factor $1/n$ handles large n), and the separation property makes F open onto its image. ∎

Since a countable basis is countably locally finite, this contains Urysohn's theorem.

6.3 Paracompactness

X is **paracompact** (Hausdorff, by convention) if every open cover has a locally finite open refinement that covers X. Compact spaces and metrizable spaces (Stone's theorem, from Lemma 6.2 with more work) are paracompact; closed subspaces of paracompact spaces are paracompact.

Theorem 6.4

(a) Every paracompact Hausdorff space is normal. (b) Every regular Lindelöf space is paracompact.

Proof. (a) First regularity: for x and closed B, cover X by $X - B$ and neighbourhoods of points of B whose closures miss x; a locally finite refinement $\mathcal{C}$ gives $V = \bigcup\{C \in \mathcal{C} : C \cap B \neq \emptyset\} \supset B$ with $x \notin \overline{V} = \bigcup \overline{C}$ (Lemma 6.1). Normality is the same argument with points replaced by a closed set. (b) A countable cover is refined to a locally finite one by subtracting closures of earlier finite unions. ■

Theorem 6.5 (Partitions of unity)

If X is paracompact Hausdorff and $\{U_\alpha\}$ is an open cover, there is a partition of unity $\{\phi_\alpha\}$ dominated by it: continuous $\phi_\alpha \geq 0$ with $\operatorname{supp}\phi_\alpha \subset U_\alpha$, the supports locally finite, and $\sum \phi_\alpha = 1$.

Proof. Shrink the cover to a locally finite cover $\{V_\alpha\}$ with $\overline{V_\alpha} \subset U_\alpha$ (paracompactness and normality), choose Urysohn functions ψ_α equal to 1 on $\overline{W_\alpha}$ (a further shrinking) and 0 off V_α, and normalize $\phi_\alpha = \psi_\alpha / \sum \psi_\beta$ (a locally finite, positive sum). ■

Partitions of unity let local constructions be glued (Riemannian metrics, imbeddings of manifolds, extensions of maps). $\mathbb{R}_\ell$ is paracompact; $\mathbb{R}_\ell^2$ and S_Ω are not; products of paracompact spaces need not be paracompact.

6.4 The Smirnov metrization theorem

A space is **locally metrizable** if every point has a metrizable neighbourhood.

Theorem 6.6 (Smirnov)

A space is metrizable iff it is paracompact Hausdorff and locally metrizable.

Proof. Cover X by open metrizable sets, take a locally finite open refinement $\mathcal{C}$, and for each C a metric basis of balls in C; for fixed radius $1/m$ the collection of $1/m$-balls in all C refined once more is locally finite. These give a countably locally finite basis; apply Nagata–Smirnov. ■

So a manifold (locally Euclidean, Hausdorff) is metrizable iff it is paracompact; the long line is a non-metrizable 1-manifold-like space because it fails paracompactness.